

Statistical and Machine Learning

Introduction

Instructions

All questions should first be discussed in pairs or small groups. Afterwards, we shall discuss them together.

Question 1

- What is machine learning (ML)?

Example answer

Machine learning develops algorithms that learn patterns from data and use them to perform tasks on new, unseen examples.

- Explain ML to a 10-year-old

Example answer

Machine learning is when we teach a computer by showing it many examples, so it can learn to do a task by itself.

- Explain ML to an undergraduate Chemistry student

Example answer

Machine learning uses data to build models that can predict outcomes or find patterns, for example in spectra, molecules, or reaction data.

- Your colleague studying Physics says that statistics and ML are old school, and we should all use AI to work with data. Is that true? What would you respond to him/her/them?

Example answer

No. AI is a broad term, and modern AI is largely built on machine learning.

Statistics is still essential because it helps us understand uncertainty, test hypotheses, design experiments, and interpret results.

A short response to the Physics colleague:

AI does not replace statistics or machine learning. It depends on them. If we want to work with data responsibly, we still need statistical thinking and ML methods.

Question 2: Iris data

We consider the `iris` data set (Example 1, lecture slides), where the goal is to classify an iris flower into one of three species based on its measured features.

1. What is the input space?

The input space consists of the measured flower features: - sepal length, - sepal width, - petal length, - petal width.

So each observation is a point in a 4-dimensional space, that is,

$$\mathcal{X} \subseteq \mathbb{R}^4.$$

2. What is the task at hand?

The task is a **supervised classification problem**.

Given the four flower measurements, we want to assign the observation to one of the three iris species: *setosa*, *versicolor*, *virginica*.

The output space consists of the three class labels,

$$\mathcal{Y} = \{\text{setosa}, \text{versicolor}, \text{virginica}\}.$$

3. What optimality criterion would you use to build the learner?

A natural criterion is to choose a classifier that makes as few wrong classifications as possible on new, unseen flowers.

In practice, this means we want good **generalization performance**, for example high classification accuracy or, equivalently, low misclassification error on test data.

4. Should the linear or non-linear classifier be used?

For the iris data, both can be considered, but a **non-linear classifier** is often more appropriate if the goal is the best predictive performance.

The reason is that the three species are not perfectly separated by a single linear boundary in the full feature space. In particular, *versicolor* and *virginica* may require more flexible decision boundaries.

Still, a **linear classifier** can work reasonably well and has the advantage of being simpler and easier to interpret.

Question 3: Image data

Consider image data from ImageNet (Example with a general image classification). Suppose the goal is to recognize whether a photo contains a cat.

1. What is the input space? Consider images in colour of sizes 32×32 pixels. What is the number of features, p , equal to?

The input space consists of digital images.

If each image has height H , width W , and C color channels, then each image can be represented as a vector in

$$\mathcal{X} \subseteq \mathbb{R}^{H \times W \times C}.$$

For example, a color image has $C = 3$ channels (red, green, blue), so each input contains a very large number of pixel values.

For a small color image of size 32×32 , we have

$$p = 32 \times 32 \times 3 = 3072.$$

2. What is the task at hand?

The task is a **supervised classification problem**.

Given an input image, we want to decide whether the image contains a cat or not.

So the output space can be written as

$$\mathcal{Y} = \{\text{cat}, \text{not cat}\}.$$

3. What optimality criterion would you use?

A natural criterion is to build a classifier that predicts correctly on new, unseen images.

In words, we want to minimize classification mistakes, or equivalently, maximize prediction accuracy on test data.

4. Why is this problem more difficult than the iris example?

This problem is much harder because the input is **high-dimensional**.

In the iris data, each observation has only 4 features.

In image data, each image may contain thousands or millions of pixel values.

This means the learner must deal with many more input variables, and therefore potentially many more model parameters.

In addition, images are more complex because:

- the same cat can appear in different positions,
- lighting can change,
- the background can vary,
- cats can have different shapes, colors, and poses.

So the relationship between input and output is much more complicated than in the iris example.

5. Why might classical statistical methods have difficulty solving this problem well?

Classical statistical algorithms often struggle in such settings because:

- the number of input variables is very large,
- the decision boundary is highly non-linear,
- raw pixels are not very informative on their own,
- important patterns depend on local structure in the image.

Traditional methods usually require carefully designed features made by humans.

By contrast, modern machine learning methods, especially deep neural networks, can learn useful features automatically from large image data sets.

6. Can you give an example of another problem where ML/AI perform exceptionally well, whereas traditional statistical methods struggle?

7. Can you give an example of another problem where ML/AI performs exceptionally well, whereas traditional statistical methods struggle?

Yes. A good example is **speech recognition**.

The input is a complex audio signal, and the task is to convert it into text. This is difficult for traditional statistical methods because the data are high-dimensional, sequential, and highly non-linear. Modern ML/AI methods, especially deep learning, can learn these complex patterns much more effectively.

Question 4: Supervised, unsupervised, and semi-supervised learning

1. Define a supervised learning problem, list a few methods useful for solving this problem, and give a data example to which one can apply supervised learning.

A **supervised learning** problem is one in which we observe inputs $\mathcal{X}$ together with labels or responses $\mathcal{Y}$, and the goal is to learn a mapping

$$f : \mathcal{X} \rightarrow \mathcal{Y}.$$

The response $\mathcal{Y}$ may be categorical, continuous, count-valued, etc.

Examples of supervised learning methods include:

- linear regression,
- logistic regression,
- k -nearest neighbors,
- decision trees,
- random forests,
- support vector machines,
- neural networks.

An example is the **iris data set**, where the input consists of flower measurements and the response is the species label.

2. Define an unsupervised learning problem, list a few methods useful for solving this problem, and give a data example to which one can apply unsupervised learning.

An **unsupervised learning** problem is one in which we observe only inputs $\mathcal{X}$, without labels or responses, and the goal is to discover structure in the data.

This may involve:

- finding clusters,
- finding a lower-dimensional representation,
- detecting anomalies,
- identifying latent structure.

A general framework is: given a data set $\mathcal{X}$, find a representation or structure that captures important patterns in the data.

Examples of unsupervised learning methods include:

- k -means clustering,
- hierarchical clustering,
- principal component analysis (PCA),
- mixture models,
- autoencoders.

An example is **gene expression data**, where one may try to group similar samples or reduce dimension to visualize patterns.

3. Can you think of a definition of semi-supervised learning?

A **semi-supervised learning** problem lies between supervised and unsupervised learning: we have a small number of labeled observations together with a larger number of unlabeled observations.

The goal is to use both kinds of data to learn a better mapping

$$f : \mathcal{X} \rightarrow \mathcal{Y}$$

than would be possible using only the labeled data.

A typical example is image classification, where labeling images is expensive, so only a small subset is labeled, while many more images are available without labels.

Question 5: Real-life case study: building a learner at Amezo

Imagine you work as a data scientist at **Amezo**, a large online retailer. You are asked to build a learner to increase the sales of cleaning products.

A possible goal is to predict which customers are most likely to buy a cleaning product after seeing a recommendation or discount.

Possible predictors include:

- past purchases,
- browsing history,
- prices and discounts,
- season,
- customer demographics.

1. How would you start solving this task?

First, I would define the business goal clearly. For example: do we want to predict whether a customer will buy a cleaning product, or do we want to rank customers by purchase probability?

Then I would identify:

- the target variable,
- the available predictors,
- the prediction time point,
- and the decision that will be taken based on the model.

After that, I would collect and clean the data, create useful features, and split the data into training and test sets.

2. What could be the input space and output space?

The input space consists of customer and product information, for example:

- past purchases,
- browsing history,
- prices,
- discounts,
- season,
- customer characteristics.

The output space could be:

- $\{0, 1\}$, where 1 means the customer buys the product,
- or a continuous outcome, such as the amount spent.

3. What is the task at hand?

This is a **supervised learning** problem.

It may be:

- a **classification problem**, if we predict whether a customer buys a product,
- or a **regression problem**, if we predict how much the customer will spend.

4. What optimality criterion would you use?

A natural criterion is good performance on new, unseen data.

Depending on the task, this may mean:

- high classification accuracy or AUC,
- low prediction error,
- or, more practically, higher sales or profit.

So the final criterion should be linked both to predictive performance and to the business goal.

5. How would you build the learner?

I would start with a simple baseline model, such as logistic regression or a decision tree, and then compare it with more flexible methods such as random forests or boosting.

I would choose variables that are relevant and available at prediction time, and I would avoid overfitting by using cross-validation or a validation set.

6. How would you test whether the learner is good?

First, I would evaluate it on a test set that was not used for training.

Then I would compare several models using suitable performance measures.

Finally, in a real business setting, I would run an **A/B test**: one group of customers would receive recommendations based on the model, and another group would not. Then I would compare the sales outcomes.

7. When can we trust and use the learner?

We can use the learner if it performs well on unseen data and if the future data are similar to the training data.

Even then, the model should be monitored over time, since customer behavior may change and the learner may need updating.